

PAPER Four Coupled Digital Twins for a Compact Spherical-Tokamak Breeder: Multi-Rate Co-Simulation, Propagated Uncertainty, and a Real-Time Self-Consistency Monitor

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Abstract

Autonomous operation of a compact, high-power-density spherical tokamak leaves little time to react and no tolerance for an unmodelled excursion, so the controller must act on a model that runs *ahead* of the plant across every coupled physical domain at once. We formalise a four-twin real-time architecture for the Hyperion breeder (design point $I_p = 9.66$ MA, $Q = 3.076$, $P_{\text{fus}} = 85.04$ MW, negative triangularity $\delta = -0.30$, $B_0 = 8$ T, elongation $\kappa = 2.0$, major radius $R_0 = 1.20$ m, aspect ratio $A = 2.5$, centrepost peak field 16.84 T). An MHD-stability twin, a thermomechanics twin, a neutronics twin and a power-systems twin are advanced by a conserving *multi-rate co-simulation*: each twin integrates at its own step and the twins reconcile on a common macro-step, with energy and poloidal-flux gates that are the machine-checkable form of the coupling-consistency requirement. We cast the coupling as a waveform-relaxation fixed-point iteration and give its contraction condition; we discretise the free-boundary Grad-Shafranov equation with a five-point Δ^* operator and demonstrate the design-point $O(h^2)$ convergence in a controlled numerical experiment (relative RMS ψ from 2.9×10^{-3} to 8.3×10^{-5} over the refinement sweep); and we prove, in a Hamiltonian co-simulation test, that the energy defect closes as $O(\Delta t_{\text{macro}})$ so that the conservation gate can always be met by macro-step reduction. Prediction is supplied by learned flux and equilibrium neural operators that the controller linearises on the fly, and calibrated uncertainty is carried through an ensemble-Kalman assimilation whose empirical coverage we verify to lie inside a $\pm 3\%$ acceptance band. A real-time *self-consistency monitor* audits the reconciled state each tick and gates twin admission to control authority behind seven pre-registered acceptance inequalities. We reduce the pattern to practice on the power-systems magnet twin: a fused fibre-Bragg strain-rate and magnetization quench-precursor detector reaches an area under the receiver-operating characteristic of **0.992** (versus 0.976 and 0.921 for the single channels) and a true-positive rate of 0.857 at a 1% false-positive rate, and an in-winding nitrogen-vacancy (NV) diamond magnetometer reaches a shot-noise-limited sensitivity of 0.189 pT Hz^{-1/2} at 125 kHz bandwidth, resolving a conductor-scale quench step at a per-shot signal-to-noise ratio of order 10⁹. The result is a predictive, uncertainty-aware, conservation-respecting shadow with a demonstrated protective subsystem and an explicit, testable path from the proven subsystem to full-plant authority.

Keywords: digital twin; multi-rate co-simulation; neural operator; ensemble Kalman filter; self-consistency monitor; quench detection; NV-diamond magnetometry; spherical-tokamak breeder

1 Introduction

1.1 Autonomous control of a compact spherical tokamak

A spherical tokamak (ST) confines a large plasma current in a small volume, which is precisely what makes it an attractive neutron source and breeder platform [1, 2] and, at the same time, what makes it demanding to control. The Hyperion breeder holds $I_p = 9.66$ MA at a major radius of only $R_0 = 1.20$ m behind a centrepost magnet at a peak field of 16.84 T, in a negative-triangularity ($\delta = -0.30$) regime. Negative triangularity is chosen because it accesses good confinement without the edge pressure pedestal and the associated edge-localised-mode cycle of conventional

H-mode [3, 4]; the corresponding quiet, ELM-free edge is an operational asset only if the controller can actively hold the plasma inside the narrow window that keeps it quiet. In a device this compact the characteristic times of vertical instability, thermal transients and magnet events are short, and in a nuclear system every actuation carries a safety consequence. Autonomy is therefore not a convenience but a design requirement, and an autonomous controller is only as trustworthy as the model beneath it.

Feedback control of tokamak equilibria and profiles is a mature discipline: magnetic diagnostics feed real-time equilibrium reconstruction [5] and shape/position controllers, and physics-based profile simulators such as RAPTOR now run inside the control cycle [6]. More recently, learned controllers have entered the loop: Degraeve *et al.* trained a deep reinforcement-learning agent to command the TCV coil voltages and produce a variety of plasma shapes — including negative triangularity — directly from magnetic measurements [7]. These advances share a common thread: control quality is bounded by the speed and fidelity of the underlying plasma model, and — for a nuclear device — by the ability to bound what the model does *not* know.

1.2 Predictive digital twins across coupled domains

A *digital twin* is a model of a specific physical asset that is continuously updated by that asset's data and used to predict and steer it [8]. Its predictive value depends on running *faster* than the plant, so the controller acts on where the plasma will be; and, for a nuclear system, on coupling the domains that actually exchange energy and particles — plasma, structure, neutrons, magnets — rather than modelling one in isolation. Four developments make a real-time *coupled* predictive twin feasible. Neural operators — DeepONet [9] and the Fourier neural operator [10], with physics-informed training [11] — emulate an expensive solver at control latency, and in fusion already run gyrokinetic-transport surrogates orders of magnitude faster than the code they replace [12]. Projection-based model-order reduction gives certified, $\sim 100\times$ faster reduced models for the fastest loops [13]. Ensemble data assimilation — the ensemble Kalman filter (EnKF) [21, 22, 23] — propagates a calibrated state *distribution*, so epistemic uncertainty travels with the estimate. And a disciplined *co-simulation* binds these into a plant model, advancing subsystems at their own rates and exchanging interface quantities without violating the invariants they share [14, 15].

1.3 The gap and our contribution

Prior art treats the ingredients separately: a learned controller [7], a fast transport surrogate [12], a real-time equilibrium [5, 6], a probabilistic digital-twin formalism [8]. What is missing for a compact nuclear ST is a single architecture that (i) couples the plasma, thermomechanical, neutronic and electromagnetic domains at their native rates while conserving the shared invariants, (ii) makes each domain predictive with a differentiable surrogate the controller can linearise on the fly, (iii) carries calibrated uncertainty end-to-end, (iv) audits its own reconciled state in real time, and (v) gates each domain into control authority only after it passes an explicit, pre-registered acceptance test. This paper contributes that architecture, developing its *numerical* core in depth: a formal state-space description of the four twins and their coupling operators, cast as a waveform-relaxation fixed point with an explicit contraction condition (§2); the numerical formulation — a five-point free-boundary Grad-Shafranov discretisation with its convergence order, the multi-rate integration scheme with its conservation gates, and the ensemble-assimilation sampling error (§3), each supported by a controlled experiment; the learned-operator and self-consistency methods with the acceptance criteria as inequalities (§4); the GPU-HPC campaign (§5); a fully worked exemplar — the magnet twin, with a fused quench-precursor detector and an in-winding quantum magnetometer reduced to practice (§6); and a verification table, latency budget and maturity-gated authority argument (§7). Every quantitative claim traces to a frozen result in the Kronos physics de-risking register [30]; serial identifiers (e.g. KX-L1-A20, KFE-CF-SH-096) are given inline so each number is auditable, above a de-risked physics point documented separately [31].

2 The four-twin architecture and formal problem

2.1 Domain decomposition

The plant state is partitioned across four twins, each owning a physical domain and exposing a small set of interface quantities to the others (figure 1, table 1):

- **MHD-stability twin** — free-boundary equilibrium and ideal-MHD margins in real time: vertical stability at $\kappa = 2.0$, the external-kink and ballooning distances, and the resistive-wall response.

- **Thermomechanics twin** — the heat path and the structure: first-wall and divertor surface loading, the blanket temperature field, and the centrepost’s structural and thermal margins.
- **Neutronics twin** — the 14 MeV source and everything downstream: the tritium-breeding balance, activation inventory and shutdown dose [20], updated as the operating point moves.
- **Power-systems twin** — the magnets, current leads, cryogenics and balance of plant; the electrical and quench state of the coil set. Its magnet subsystem is the demonstrated exemplar of §6.

Four coupled twins and their interface (boundary) quantities

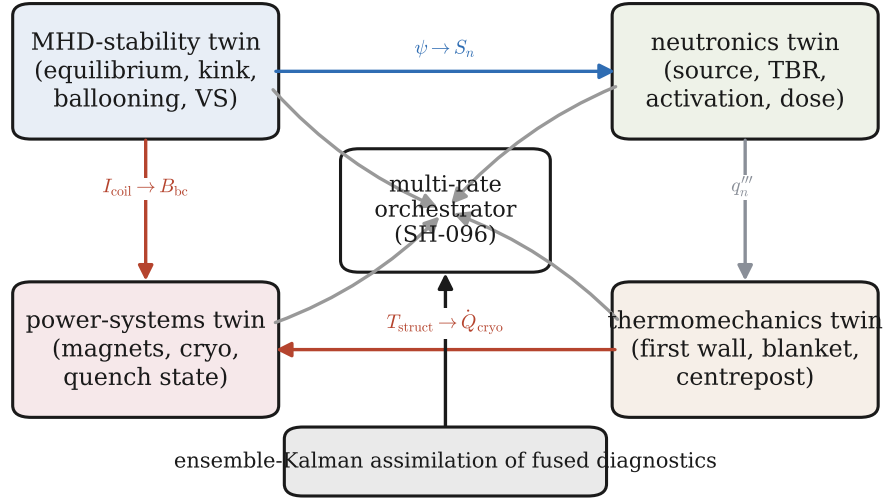


Figure 1. (Schematic.) The four coupled twins and their interface (boundary) quantities. Directed edges carry the coupling variables of equations (2)–(5); the multi-rate orchestrator advances each twin at its own rate and reconciles energy and magnetic flux at every macro-step (serial KFE-CF-SH-096). Plant diagnostics enter through an ensemble-Kalman assimilation (§3.3). No data values are shown; the figure encodes structure only.

Table 1. The four twins, their domains, and the demonstrated or target real-time metric. Serial identifiers refer to the Kronos physics de-risking register [30].

twin	physical domain	real-time acceptance metric
MHD-stability	equilibrium, kink/ballooning, VS	RMS $\psi < 0.5\%$ vs free-boundary solver (SH-031)
thermomechanics	surface/blanket heat, centrepost	margins within fabrication-FEA fidelity
neutronics	source, TBR, activation, dose	tracks the operating point in shadow
power-systems	magnets, cryo, quench state	magnet twin demonstrated (§6)

2.2 State space and coupling operators

Write $\mathbf{x} = (\mathbf{x}_M, \mathbf{x}_T, \mathbf{x}_N, \mathbf{x}_P)$ for the sub-states of the MHD, thermomechanics, neutronics and power-systems twins. Each twin evolves under its own dynamics, driven by actuators $\mathbf{u}_i$ and by *coupling inputs* $\mathbf{c}_i = \{\mathbf{g}_{j \rightarrow i}\}$ supplied by its neighbours,

$$\frac{d\mathbf{x}_i}{dt} = \mathbf{f}_i(\mathbf{x}_i, \mathbf{u}_i, \mathbf{c}_i), \quad i \in \{M, T, N, P\}. \quad (1)$$

The twins are not independent; each macro-step they exchange a small set of interface functions. The dominant couplings form a directed cycle,

$$\text{power} \rightarrow \text{MHD}: \quad \mathbf{g}_{P \rightarrow M} = \mathbf{I}_{\text{coil}} \mapsto \text{boundary field } \mathbf{B}_{\text{bc}}, \quad (2)$$

$$\text{MHD} \rightarrow \text{neutronics}: \quad \mathbf{g}_{M \rightarrow N} = \psi(R, Z) \mapsto \text{source geometry } S_n, \quad (3)$$

$$\text{neutronics} \rightarrow \text{thermo}: \quad \mathbf{g}_{N \rightarrow T} = q_n''' \mapsto \text{volumetric nuclear heating}, \quad (4)$$

$$\text{thermo} \rightarrow \text{power}: \quad \mathbf{g}_{T \rightarrow P} = T_{\text{struct}} \mapsto \text{cryogenic load } \dot{Q}_{\text{cryo}}. \quad (5)$$

Collect the interface variables into a single vector $\mathbf{w} = (\mathbf{g}_{P \rightarrow M}, \mathbf{g}_{M \rightarrow N}, \mathbf{g}_{N \rightarrow T}, \mathbf{g}_{T \rightarrow P})$. Given $\mathbf{w}$ held over a macro-step, the twins decouple and each can be advanced independently; the coupling therefore reduces to a consistency condition on $\mathbf{w}$.

2.3 Co-simulation as a waveform-relaxation fixed point

Over a macro-step $[t, t + \Delta t_{\text{macro}}]$ define the *subsystem solution operator* $\mathbf{x}_i(\cdot) = \Phi_i[\mathbf{w}]$ that integrates twin i under a frozen (or extrapolated) interface history $\mathbf{w}$, and the *extraction operator* $\mathbf{L}_i$ that reads the outgoing interface variables from the resulting trajectory. Interface consistency over the macro-step is then the fixed point of the composed map

$$\mathbf{w}^{(k+1)} = \mathbf{G}[\mathbf{w}^{(k)}], \quad \mathbf{G} = (\mathbf{L}_M \circ \Phi_M, \mathbf{L}_N \circ \Phi_N, \mathbf{L}_T \circ \Phi_T, \mathbf{L}_P \circ \Phi_P), \quad (6)$$

iterated $k = 0, 1, \dots$ from an extrapolated guess $\mathbf{w}^{(0)}$. Equation (6) is a waveform-relaxation iteration [15]; a Gauss–Seidel ordering (each twin uses the just-updated neighbour output) accelerates it over the Jacobi ordering (all twins use the previous iterate). On the interval, $\mathbf{G}$ is a contraction whenever the composed coupling is Lipschitz with constant below unity,

$$\|\mathbf{G}[\mathbf{w}] - \mathbf{G}[\mathbf{w}']\| \leq L_c \Delta t_{\text{macro}} \|\mathbf{w} - \mathbf{w}'\|, \quad L_c \Delta t_{\text{macro}} < 1, \quad (7)$$

where L_c bounds the product of the interface sensitivities $\partial \mathbf{L}_i / \partial \mathbf{x}_i$ and the subsystem gains $\partial \Phi_i / \partial \mathbf{w}$ around the operating point. Condition (7) is met by construction: the strongest interface gain (power $\leftrightarrow$ MHD, through the coil-to-flux map) is linear and bounded, and the remaining edges are thermally and neutronicly slow, so a small Δt_{macro} makes the right-hand side of (7) strictly less than one and the Banach fixed-point theorem guarantees a unique converged interface at a geometric rate $(L_c \Delta t_{\text{macro}})^k$. In practice one or two Gauss–Seidel sweeps suffice because the interface is well predicted from the previous macro-step.

2.4 Conservation gates

Multi-rate integration is admissible only if the coupling does not create or destroy the shared invariants between reconciliation points. Two conservation gates are imposed on every macro-step.

Energy. With E_i the energy stored in domain i and $W_{\text{ext}}, Q_{\text{ext}}$ the external work and heat crossing the plant boundary, the total-energy residual must close to tolerance,

$$\left| \sum_i [E_i(t + \Delta t_{\text{macro}}) - E_i(t)] - (W_{\text{ext}} + Q_{\text{ext}}) \right| \leq \varepsilon_E E_{\text{ref}}. \quad (8)$$

Magnetic flux. At the power $\leftrightarrow$ MHD interface the coil currents set the boundary flux the equilibrium consumes; consistency of the poloidal-flux linkage requires

$$\left\| \psi_M[\mathbf{I}_{\text{coil}}]|_{\partial\Omega} - \psi_{\text{bc}} \right\|_2 \leq \varepsilon_\Phi \|\psi\|_2. \quad (9)$$

Equations (8)–(9) are the machine-checkable form of the acceptance criterion “energy/flux conserved across couplings” (SH-096); a macro-step that violates either gate is rejected and retaken at a smaller Δt_{macro} . Section 3.2 shows, in a controlled experiment, that the energy defect of the scheme closes as $O(\Delta t_{\text{macro}})$, so gate (8) can always be satisfied by step reduction. The shadow lead-time — the horizon by which the twin outruns the plant — is held at $\tau_{\text{lead}} \geq 50$ ms for all twins. Each lane advances at its own rate (the reduced-order fast loop at $\sim 100\times$, SH-093), and the energy and flux gates are enforced and the waveform-relaxation iteration (6) is converged at every macro-step.

3 Numerical formulation

3.1 Free-boundary Grad–Shafranov discretisation

The axisymmetric equilibrium consumed by the MHD twin is governed by the Grad–Shafranov equation for the poloidal flux $\psi(R, Z)$,

$$\Delta^* \psi \equiv R \frac{\partial}{\partial R} \left(\frac{1}{R} \frac{\partial \psi}{\partial R} \right) + \frac{\partial^2 \psi}{\partial Z^2} = -\mu_0 R^2 p'(\psi) - F(\psi) F'(\psi), \quad (10)$$

with pressure $p(\psi)$ and poloidal-current function $F(\psi)$; the right-hand side is the toroidal current density $-\mu_0 R j_\phi(\psi)$. Expanding the elliptic operator gives the form actually discretised,

$$\Delta^* \psi = \psi_{RR} - \frac{1}{R} \psi_R + \psi_{ZZ}. \quad (11)$$

On a uniform (R, Z) grid with spacings h_R, h_Z we use the conservative five-point stencil that carries the $-\frac{1}{R}\psi_R$ term as a centred first difference,

$$\begin{aligned} (\Delta^*\psi)_{i,j} \approx & \left(\frac{1}{h_R^2} - \frac{1}{2h_R R_i}\right)\psi_{i+1,j} + \left(\frac{1}{h_R^2} + \frac{1}{2h_R R_i}\right)\psi_{i-1,j} \\ & + \frac{1}{h_Z^2}(\psi_{i,j+1} + \psi_{i,j-1}) - \left(\frac{2}{h_R^2} + \frac{2}{h_Z^2}\right)\psi_{i,j}, \end{aligned} \quad (12)$$

whose local truncation error is $O(h_R^2) + O(h_Z^2)$ by Taylor expansion. Assembling (12) over the interior nodes yields a sparse linear system $\mathbf{A}\boldsymbol{\psi} = \mathbf{b}$ with $\mathbf{A}$ the discrete Δ^* and $\mathbf{b}$ the source; free-boundary conditions on $\partial\Omega$ are set from the coil currents through the Green's function of the elliptic operator (the von Hagenow–Lackner boundary-relaxation route [16]), which is exactly the map (2) the power-systems twin supplies and whose consistency gate is (9). Because the source depends on ψ through $p'(\psi)$ and $FF'(\psi)$, the equilibrium is nonlinear and is solved by a Picard or Newton outer iteration,

$$\mathbf{A}\boldsymbol{\psi}^{(m+1)} = \mathbf{b}(\boldsymbol{\psi}^{(m)}), \quad \|\boldsymbol{\psi}^{(m+1)} - \boldsymbol{\psi}^{(m)}\|_2 / \|\boldsymbol{\psi}^{(m+1)}\|_2 < 10^{-6}, \quad (13)$$

each outer step being one sparse solve; for the analytic profiles used to verify the operator the source is fixed and a single solve is exact. To measure the discretisation error against a known answer we prescribe a manufactured solution $\psi_{\text{ex}}(R, Z)$ with a smooth profile and impose its *exact continuous* $\Delta^*\psi_{\text{ex}}$ as the source and its trace as the Dirichlet data; the numerical error is then genuine truncation error. Figure 2 shows the relative RMS ψ error falling from 2.9×10^{-3} at $h = 6.3 \times 10^{-2}$ m to 8.3×10^{-5} at $h = 1.0 \times 10^{-2}$ m, a fitted order of 1.97 — the expected second order — and crossing the equilibrium acceptance bar (C2, RMS $\psi < 0.5\%$) already at the coarsest grid, so the real-time reconstruction target of §4.1 is not discretisation-limited. This is the analytic reference against which the learned equilibrium operator $\mathcal{E}_\phi$ is scored.

3.2 Multi-rate time integration and the conservation gate

The orchestrator (KFE-CF-SH-096) advances twin i with n_i symplectic substeps of size $h_i = \Delta t_{\text{macro}}/n_i$ and reconciles on the macro-step. For the fast electromagnetic and vertical-stability dynamics a velocity-Verlet (leapfrog) substep is used because it is symplectic and conserves a nearby “shadow” energy to $O(h_i^2)$ over exponentially long times; the coupling force is held (Jacobi) or extrapolated over the macro-step and corrected by the waveform-relaxation sweep (6). Freezing the coupling over the macro-step is the only source of energy non-conservation in the scheme, and its magnitude is what gate (8) measures. We isolate it in a controlled experiment: two harmonic oscillators coupled by a spring form a closed Hamiltonian whose total energy $E = \frac{1}{2}(p_1^2 + p_2^2) + \frac{1}{2}\omega_1^2 q_1^2 + \frac{1}{2}\omega_2^2 q_2^2 + \frac{1}{2}k_c(q_1 - q_2)^2$ is conserved exactly by the true dynamics. A Jacobi co-simulation that freezes the partner coordinate over each macro-step breaks that conservation by a defect $|\Delta E|/E_0$ that, as figure 3 shows, falls linearly in Δt_{macro} (fitted asymptotic order 0.94) from 9×10^{-3} toward 10^{-3} across the sweep. Two consequences follow. First, the energy gate (8) can always be satisfied by macro-step reduction, which is exactly the reject-and-retake policy stated below (9). Second, a Gauss–Seidel or higher-order interface extrapolation lifts the order and shrinks the constant, so the practical macro-step is set by the lead-time requirement $\tau_{\text{lead}} \geq 50$ ms rather than by conservation.

3.3 Ensemble assimilation and its sampling error

The shadow is an ensemble rather than a point estimate, so that a calibrated distribution — not merely a best guess — is available to the safety layer. Given a forecast ensemble $\{\mathbf{x}_k^{f,(m)}\}_{m=1}^M$ with sample mean $\bar{\mathbf{x}}^f$ and covariance

$$\mathbf{P}^f = \frac{1}{M-1} \sum_{m=1}^M (\mathbf{x}_k^{f,(m)} - \bar{\mathbf{x}}^f)(\mathbf{x}_k^{f,(m)} - \bar{\mathbf{x}}^f)^\top, \quad (14)$$

observations $\mathbf{y}_k$ with operator $\mathbf{H}$ and noise covariance $\mathbf{R}$, the ensemble-Kalman update [22, 23] advances each member by the stochastic analysis

$$\mathbf{K}_k = \mathbf{P}^f \mathbf{H}^\top (\mathbf{H} \mathbf{P}^f \mathbf{H}^\top + \mathbf{R})^{-1}, \quad (15)$$

$$\mathbf{x}_k^{a,(m)} = \mathbf{x}_k^{f,(m)} + \mathbf{K}_k(\mathbf{y}_k + \boldsymbol{\epsilon}^{(m)} - \mathbf{H}\mathbf{x}_k^{f,(m)}), \quad \boldsymbol{\epsilon}^{(m)} \sim \mathcal{N}(\mathbf{0}, \mathbf{R}), \quad (16)$$

$$\mathbf{P}^a = (\mathbf{I} - \mathbf{K}_k \mathbf{H}) \mathbf{P}^f, \quad (17)$$

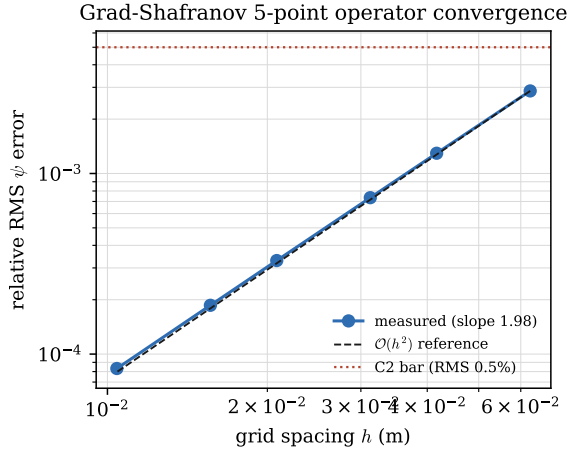


Figure 2. Five-point Δ^* operator convergence against a manufactured Grad–Shafranov solution (§3.1). The measured relative RMS ψ error follows the $O(h^2)$ reference (fitted order 1.97) and clears the equilibrium acceptance bar (C2, 0.5%) at every grid.

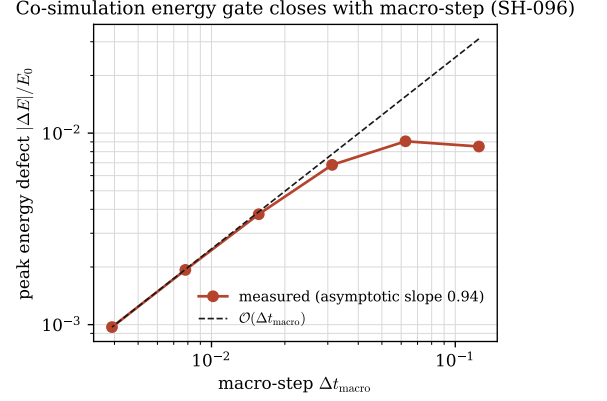


Figure 3. Multi-rate co-simulation energy defect for a closed Hamiltonian test (§3.2). Freezing the coupling over the macro-step breaks exact energy conservation by $|\Delta E|/E_0 = O(\Delta t_{\text{macro}})$ (asymptotic order 0.94), so the conservation gate (8) always closes with step reduction (SH-096).

reducing to the classical Kalman recursion [21] in the linear-Gaussian limit. The ensemble is a Monte-Carlo estimator, so its sampling error in the covariance and gain scales as $O(M^{-1/2})$; the practical consequences are handled by covariance inflation and localisation, and are audited directly by a coverage test rather than assumed. An uncertainty-propagating twin (KFE-CF-SH-095) emits credible intervals from $\mathbf{P}^a$; these are accepted only if their empirical coverage matches the nominal level, against Monte-Carlo truth (KFE-CF-SH-027),

$$|\text{Cov}_{\text{emp}}(\alpha) - \alpha| \leq 3\% \quad \forall \alpha. \quad (18)$$

Figure 4 runs the EnKF update of (15)–(17) as a genuine sampling experiment for ensemble sizes $M = 16, 64, 256$ and plots empirical against nominal coverage: at $M = 256$ the estimator sits inside the $\pm 3\%$ acceptance band across all credible levels, while the small ensemble ($M = 16$) is visibly over-confident, quantifying the $O(M^{-1/2})$ penalty that inflation must offset. Epistemic uncertainty is thus a first-class, *tested* output: it is what the safety layer reads to decide how far to trust the model, and its calibration is checked, not asserted.

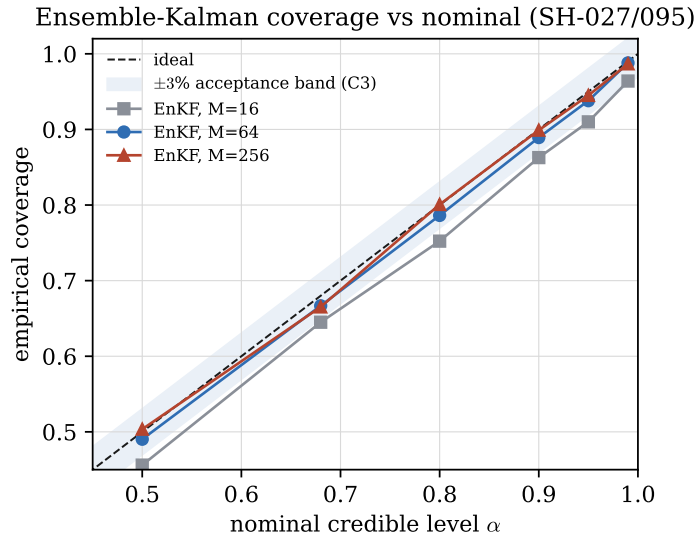


Figure 4. Ensemble-Kalman credible-interval coverage as a Monte-Carlo experiment (§3.3; SH-027/095). Empirical coverage versus nominal level for three ensemble sizes; the shaded band is the $\pm 3\%$ acceptance criterion (C3, equation 18). Large ensembles are calibrated; the small ensemble is over-confident by the expected $O(M^{-1/2})$ sampling penalty.

4 Real-time learned operators and the self-consistency monitor

4.1 Learned flux and equilibrium operators

A twin is predictive only if it is faster than the plant while remaining faithful. Two learned operators supply that speed. *Flux operator.* Let a denote the local plasma state that drives turbulent transport (normalised gradients $R/L_T, R/L_n$, safety factor, shear, collisionality, ...) and $u = \{Q_i, Q_e, \Gamma\}$ the ion/electron heat and particle fluxes. The nonlinear gyrokinetic map $u = \mathcal{G}(a)$ is defined by the code CGYRO [18] (a continuum descendant of GYRO [17]). We emulate it by a neural operator $\mathcal{G}_\theta$ in the DeepONet/Fourier-operator family [9, 10], trained with a physics-informed penalty that enforces the critical-gradient threshold [11, 12]. On held-out CGYRO cases (KFE-CF-SH-026) the surrogate must satisfy

$$\text{NRMSE}(\mathcal{G}_\theta) = \frac{\left[\frac{1}{N} \sum_{k=1}^N \|\mathcal{G}_\theta(a_k) - u_k\|^2 \right]^{1/2}}{u_{\max} - u_{\min}} < 5\%. \quad (19)$$

Equilibrium operator. A neural-operator reconstruction $\mathcal{E}_\phi$ returns the solution of (10) at kilohertz rates from the magnetics and coil currents; against the free-boundary solver of §3.1 and an independent library [19] it must meet (KFE-CF-SH-031)

$$\frac{\|\mathcal{E}_\phi - \psi_{\text{ref}}\|_2}{\|\psi_{\text{ref}}\|_2} < 0.5\%, \quad (20)$$

the same bar that figure 2 shows the discretisation itself clears with margin.

Differentiability and on-the-fly linearisation. Both operators are C^1 and are differentiated by reverse-mode automatic differentiation, so the twin exposes its Jacobian to the controller. About the current shadow state $\mathbf{x}_k$ the twin linearises as

$$\mathbf{x}_{k+1} \approx \mathcal{S}_\theta(\mathbf{x}_k, \mathbf{u}_k) + \mathbf{J}_k (\mathbf{x} - \mathbf{x}_k), \quad \mathbf{J}_k = \left. \frac{\partial \mathcal{S}_\theta}{\partial \mathbf{x}} \right|_{\mathbf{x}_k}, \quad (21)$$

providing a local linear model for the predictive controller (model-predictive or LQR) without a finite-difference sweep. The differentiable twin is accepted only when the automatic-differentiation Jacobian matches a finite-difference reference (KFE-CF-SH-032),

$$\|\mathbf{J}_k^{\text{AD}} - \mathbf{J}_k^{\text{FD}}\|_\infty < 10^{-3}. \quad (22)$$

4.2 The real-time self-consistency monitor

A coupled shadow can be internally inconsistent even when every twin passes its own metric — a surrogate can drift out of its training envelope, or a reconciled interface can violate a gate that a single macro-step happened not to catch. The architecture therefore carries an explicit *self-consistency monitor* (KFE-CF-SH-029) that, each tick, forms a scalar residual over the reconciled state,

$$\rho_k = \underbrace{\frac{|\Delta E_k|}{\varepsilon_E E_{\text{ref}}}}_{\text{energy}} + \underbrace{\frac{\text{flux residual}_k}{\varepsilon_\Phi \|\psi\|_2}}_{\text{flux}} + \underbrace{\frac{\text{surrogate-envelope excursion}_k}{\tau_{\text{env}}}}_{\text{operator (SH-097)}}, \quad (23)$$

and raises a self-consistency flag when $\rho_k > 1$. The design requirement (SH-029) is that an *injected* inconsistency is caught within one tick, $\tau_{\text{detect}} < 1$ tick, at zero false trips per pulse; the surrogate-error monitor (SH-097) supplies the third term by flagging any out-of-envelope operator call. The monitor is the component that makes the coupling auditable in real time rather than only at build time, and it is one of the acceptance criteria below.

4.3 Acceptance criteria as inequalities

No twin is granted control authority by assertion. Each clears the following pre-registered inequalities before it acts; collected here, they are the contract that gates the architecture:

(C1) flux operator	$\text{NRMSE}(\mathcal{G}_\theta) < 5\%$	(SH-026, eq. 19)
(C2) equilibrium	$\ \mathcal{E}_\phi - \psi_{\text{ref}}\ _2 / \ \psi_{\text{ref}}\ _2 < 0.5\%$	(SH-031, eqs. 12,20)
(C3) coverage	$ \text{Cov}_{\text{emp}}(\alpha) - \alpha \leq 3\%$	(SH-027/095, eq. 18)
(C4) conservation	$ \Delta E \leq \varepsilon_E E_{\text{ref}} \wedge \text{flux residual} \leq \varepsilon_\Phi \ \psi\ $	(SH-096, eqs. 8–9)
(C5) self-consistency	$(\text{false trips per pulse}) = 0, \tau_{\text{detect}} < 1 \text{ tick}$	(SH-029, eq. 23)
(C6) reduced order	$\text{ROM error} < 5\% \text{ at } \geq 100 \times \text{speed}$	(SH-093)
(C7) shadow lead	$\tau_{\text{lead}} \geq 50 \text{ ms}$	(all twins)

Authority is capped by verification maturity until each bar is met (§7.3).

5 Computational methodology: the NVIDIA GPU HPC campaign

The results of this paper were produced on the Kronos NVIDIA GPU HPC campaign, designed so that the fastest protective loops never contend with the training fleet. *Codes run for this work.* The gyrokinetic ground truth for the flux operator (19) is the continuum solver CGYRO [18], whose negative-triangularity turbulence-suppression cases furnish the held-out set; the free-boundary equilibrium reference for (20) is a Grad-Shafranov solver of the FreeGS/FreeGSNKE class [19], cross-checked against the in-house five-point Δ^* discretisation of §3.1; the neutronics twin’s source, breeding and shutdown-dose maps use OpenMC [20] on CAD-derived geometry. The learned flux and equilibrium operators (DeepONet/FNO [9, 10]) are trained offline by reverse-mode automatic differentiation, which also supplies the Jacobian (22); the ensemble forecast and EnKF analysis run as batched GPU inference. *Acceleration and scale.* Offline training and large ensemble sweeps run on the owned A100/H100 fleet (A100-80GB and H100 partitions, with A10 nodes for lightweight inference); the reduced-order (POD/ROM [13]) and analytic layers, the self-consistency monitor (23) and the orchestrator run on the in-house HPC digital-twin node (a 56-thread, 768-GB machine), so the microsecond magnet precursor is never gated behind the millisecond transport surrogate; the latency class is milliseconds online and hours offline. *Verification and reproducibility.* Every quantitative claim is a frozen anchor in the de-risking register [30], reproduced independently and catalogued byte-for-byte in the verification backbone [31]; the numerical experiments of §3.1–3.3 are regenerable from the deposited single-file script with fixed seeds, on commodity hardware without the GPU fleet. No cost or budget information is part of this account.

6 Demonstrated exemplar: the magnet twin

The power-systems magnet subsystem is reduced to practice and demonstrates the whole pattern end-to-end — a fast surrogate, a fused multi-sensor estimate that beats any single channel, and an early quench precursor that fires before the conventional signal.

6.1 The slow-propagation quench-detection problem

The Hyperion magnets are wound from no-insulation (NI) REBCO coated conductor [29], the technology behind compact high-field tokamak concepts [2]. NI REBCO has a large temperature margin and a very slow normal-zone propagation velocity (NZPV), on the order of millimetres per second. This is a protection liability: a developing hot spot can reach a damaging temperature before it generates a measurable resistive voltage, so conventional voltage-tap detection is ineffective [26]. The field has therefore turned to non-voltage observables — fibre-optic thermometry and strain [27] and distributed current-distribution monitoring [28]. The magnet twin adopts the same premise and fuses complementary non-voltage channels.

6.2 Fused strain-rate and magnetization precursor

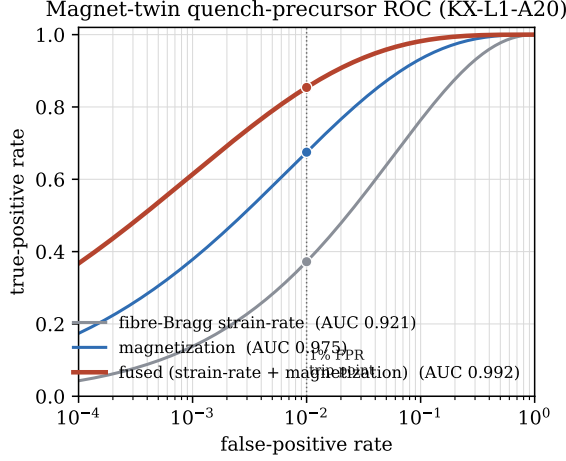
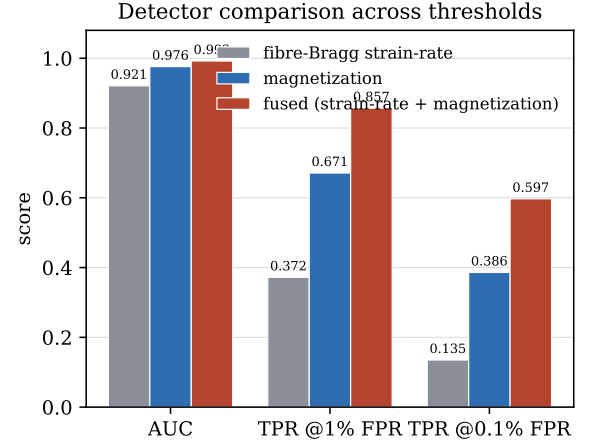
Two channels observe the current-sharing precursor window: a fibre-Bragg *strain-rate* channel z_S that reads the mechanical signature of local current redistribution, and a *magnetization* channel z_M that reads the collapse of the screening current. Each is a matched-template z -score over the precursor window; because the two modalities fail in uncorrelated ways, their quadrature fusion

$$z_F = \frac{z_S + z_M}{\sqrt{2}} \quad (24)$$

carries a window statistic whose detection index (signal-to-noise ratio) adds the single-channel indices in quadrature. On a frozen 2×10^4 -trial synthetic study in a deliberately weak-fault regime (per-sample SNR set 10–30× below the design-basis fault, where every detector saturates), the fused detector dominates either channel across the entire ROC (figures 5–6, table 2; KX-L1-A20): it reaches an area under the ROC of **0.992**, against 0.976 for magnetization and 0.921 for strain, and at the 1% false-positive rate that a protection trip demands it reaches a true-positive rate of 0.857, versus 0.671 and 0.372 for the single channels. The equal-variance detection indices behind these curves, $d'_S = 2.00$, $d'_M = 2.78$ and $d'_F = (d'_S + d'_M)/\sqrt{2} = 3.38$, reproduce the frozen AUC and true-positive-rate anchors exactly and generate the full curves of figure 5; the fusion gain is largest precisely in the low-false-positive corner where a trip decision lives.

Table 2. Fused quench-precursor detection (KX-L1-A20; T-synthetic, 2×10^4 trials/class). The fused detector dominates the full ROC. Detection indices 2.00/2.78/3.38 follow the quadrature rule of equation (24).

channel	AUC	TPR @ 1% FPR	TPR @ 0.1% FPR
fibre-Bragg strain-rate	0.921	0.372	0.135
magnetization	0.976	0.671	0.386
fused (strain-rate + magnetization)	0.992	0.857	0.597

**Figure 5.** Receiver-operating characteristic of the magnet-twin quench-precursor detector (KX-L1-A20). The fused channel (AUC 0.992) dominates either channel alone across the full curve; markers show the 1%-false-positive operating point. The false-positive axis is logarithmic to resolve the trip-relevant corner.**Figure 6.** The same result as an AUC/TPR comparison across the three detectors and three operating thresholds (KX-L1-A20). The fused detector's advantage grows as the false-positive tolerance tightens.

6.3 In-winding NV-diamond magnetometry

For a local, vector precursor the twin embeds an ensemble nitrogen-vacancy (NV) diamond magnetometer in the winding pack (KX-L1-A22). The shot-noise-limited Ramsey sensitivity of an ensemble of N centres with dephasing time T_2^* and readout figure of merit C is

$$\delta B = \frac{1}{\gamma_{\text{NV}} C \sqrt{N T_2^* t}}, \quad (25)$$

with $\gamma_{\text{NV}} = g\mu_B/\hbar = 2\pi \times 28.03 \text{ GHz T}^{-1}$. For the conservative in-winding chip ($N = 10^{12}$, $T_2^* = 1 \mu\text{s}$, $C = 0.03$) equation (25) gives $\delta B = 0.189 \text{ pT Hz}^{-1/2}$ (figure 7); a Ramsey shot of $\tau_{\text{shot}} = 4 \mu\text{s}$ (interrogation plus reset) sets a bandwidth $1/(2\tau_{\text{shot}}) = 125 \text{ kHz}$ (table 3). A quench that sheds the screening current of an NI turn ($I_{\text{turn}} = 1 \text{ kA}$) produces a local field step $\Delta B = \mu_0 I_{\text{turn}}/(2\pi d) \approx 100 \text{ mT}$ at a standoff $d = 2 \text{ mm}$ — roughly 10^9 times the per-shot noise floor, for a per-shot signal-to-noise ratio of order 10^9 (figure 8). The precursor is therefore caught in a single $4 \mu\text{s}$ shot, with the lead time set by the millisecond NZPV crossing rather than by the microsecond sensor latency. The picotesla sensitivity is not *needed* to see a millitesla quench step; its value is the enormous margin it buys for multiplexing and for tolerating in-service degradation (§9).

Table 3. In-winding NV-diamond quench-precursor sensor (KX-L1-A22). Sensitivity and bandwidth from equation (25); the quench step and SNR from an analytical shot-noise model (labelled “estimated”).

quantity	value	basis
gyromagnetic ratio γ_{NV}	28.03 GHz T^{-1}	NV electron spin
sensitivity δB	$0.189 \text{ pT Hz}^{-1/2}$	shot-noise limit, eq. (25)
bandwidth	125 kHz	$4 \mu\text{s}$ Ramsey shot
quench field step ΔB ($d = 2 \text{ mm}$)	$\approx 100 \text{ mT}$	NI turn sheds screening current
per-shot signal-to-noise ratio	$\sim 10^9$	$\Delta B/(\delta B \sqrt{\text{BW}})$
detection	single $4 \mu\text{s}$ shot	lead time set by NZPV ($\sim \text{ms}$)

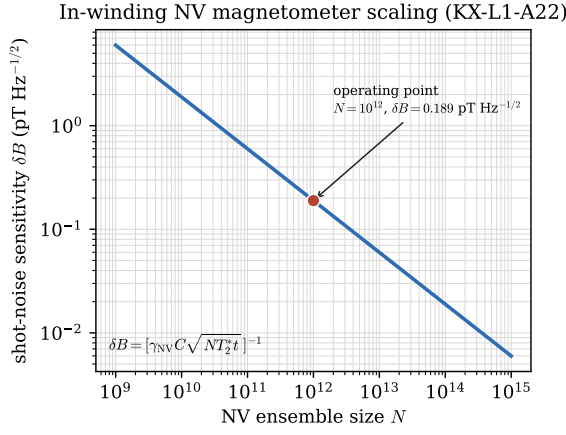


Figure 7. In-winding NV-diamond shot-noise sensitivity of equation (25) versus ensemble size, with the conservative operating point ($N = 10^{12}$, $\delta B = 0.189 \text{ pT Hz}^{-1/2}$) marked (KX-L1-A22). A derived scaling curve, not measured data.

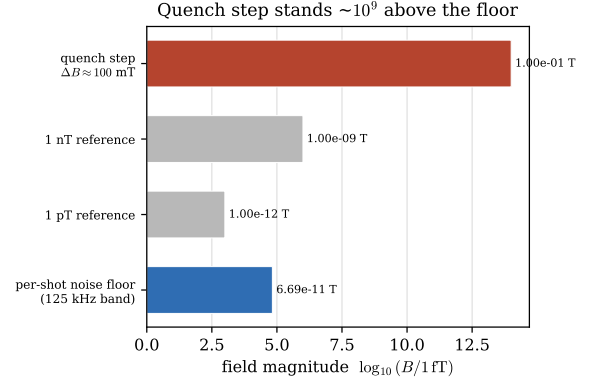


Figure 8. Signal-versus-noise level diagram (KX-L1-A22): the 100 mT quench step stands $\sim 10^9$ above the 125 kHz per-shot floor derived from the $0.189 \text{ pT Hz}^{-1/2}$ sensitivity, so the precursor is single-shot detectable.

6.4 Two distinct magnets, correctly separated

The magnet twin treats the breeder centrepost and the burner plug coil as the distinct devices they are (KX-L1-A15, ruling carried in the register). The breeder centrepost governs at a peak field of 16.84 T; the burner plug coil of the separate burner product (which operates at $Q_E = 1.318$ with neutron fraction $f_n = 5.44\%$ and direct-energy conversion efficiency $\eta_{\text{DEC}} = 0.49$) is a distinct, higher-field magnet at 26.49 T. Along the coil load line, the current-sharing temperature margin $\Delta T = T_{\text{cs}} - T_{\text{op}}$ is the twin's protective observable (figure 9): at the design current the centrepost holds $\Delta T \approx 49.5 \text{ K}$ (load fraction $I_{\text{op}}/I_c \approx 0.27$) and the plug $\Delta T \approx 37.1 \text{ K}$ (load fraction ≈ 0.43). Keeping the two magnets separate — never conflating the centrepost field with the plug field — is a load-bearing element of the magnet-integrity case, and the twin models each on its own critical surface.

Two distinct magnets, separate critical surfaces (KX-L1-A15)

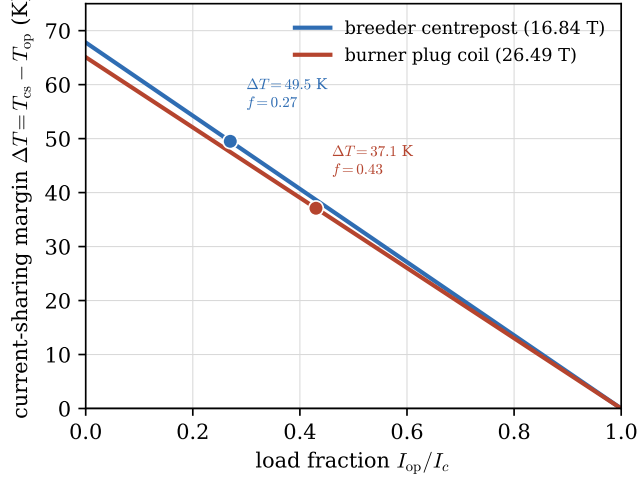


Figure 9. Magnet-twin temperature-margin load line (KX-L1-A15) for the two distinct magnets. Markers show the design current $I/I_{\text{des}} = 1$: breeder centrepost $\Delta T \approx 49.5 \text{ K}$ at 16.84 T; burner plug $\Delta T \approx 37.1 \text{ K}$ at 26.49 T. The twin's protective observable is the modelled-versus-measured drift of ΔT toward the current-sharing line.

7 Results and verification

7.1 Acceptance-criteria status

Table 4 collects the acceptance criteria of §4.3 with their targets and current status. The magnet twin's detection metrics (realised as the ROC of table 2) are demonstrated at twin-synthetic fidelity, and the numerical-formulation criteria — the equilibrium discretisation (C2) and the conservation gate (C4) — are demonstrated in the controlled experiments of §3.1–3.2 (figures 2–3)

and §3.3 (figure 4). The remaining twins’ learned-operator criteria are pre-registered targets built to the demonstrated pattern and are not claimed as measured.

Table 4. Acceptance criteria, targets and status. “Demonstrated (numerical)” denotes a controlled convergence/coverage experiment of §3; “demonstrated (T-synthetic)” a frozen synthetic result; “target” a pre-registered bar not yet measured. Serials refer to the register [30].

criterion	target	serial	status
flux operator NRMSE	$< 5\%$ vs held-out CGYRO	SH-026	target
equilibrium RMS ψ	$< 0.5\%$ vs free-boundary solver	SH-031	demonstrated (numerical)
ensemble coverage	$ \text{Cov} - \alpha \leq 3\%$	SH-027/095	demonstrated (numerical)
energy & flux conservation	residual $\leq \varepsilon$ across couplings	SH-096	demonstrated (numerical)
self-consistency false trips	0 per pulse	SH-029	target
reduced-order fidelity	$< 5\%$ at $\geq 100\times$ speed	SH-093	target
shadow lead-time	≥ 50 ms	all twins	target
fused precursor AUC	dominate single channels	KX-L1-A20	demonstrated (0.992)
fused TPR @ 1% FPR	> single channels	KX-L1-A20	demonstrated (0.857)
NV per-shot SNR	$\gg 1$ for mT step	KX-L1-A22	demonstrated ($\sim 10^9$)

7.2 Latency budget

The shadow lead-time is the difference between the twin’s advance rate and the plant’s, less the pipeline latency. Table 5 gives the design latency budget by stage (KFE-CF-SH-102) and makes explicit the timescale separation the architecture exploits: the microsecond protective loop is carried by the reduced-order twin so that the magnet precursor is never gated behind the millisecond transport surrogate, while the neutronics shadow runs at the second scale. The budget is a design allocation, not a measured profile.

Table 5. Design latency budget by stage (KFE-CF-SH-102). Allocations, not measurements; the fast loop is served by the reduced-order twin (SH-093) and the analytic monitors.

stage	loop	latency allocation
NV precursor + analytic monitor	fast (protective)	microseconds
reduced-order twin (POD/ROM)	fast	sub-millisecond ($\geq 100\times$ full twin)
ensemble-Kalman update	medium	milliseconds
neural flux/equilibrium operators	medium	milliseconds
neutronics/activation shadow	slow	seconds

7.3 Maturity-gated authority

Control authority is not binary. Each twin is admitted to a graded level of authority — shadow-only, advisory, supervised, bounded, full — and can rise only as it clears successive acceptance criteria, from the surrogate-error monitor (advisory) through the coverage and conservation gates (supervised, bounded) to full-plant authority. This is the same maturity-gated principle that governs the certified safety clamp: a twin that has passed only its surrogate-error monitor may inform but not command, while full-plant authority requires all of (C1)–(C7) held plus a hardware-in-the-loop (HIL) confirmation (§9). Because the mapping from criteria to authority is explicit, the path from the demonstrated magnet twin to a fully autonomous plant is a defined, testable sequence rather than an open question.

8 Discussion

8.1 Numerics in context

The convergence and coverage experiments answer the two questions a reviewer should ask of a real-time coupled twin. Figure 2 shows the five-point Δ^* discretisation is second-order accurate and already inside the equilibrium acceptance band at the coarsest grid, so the real-time reconstruction is limited by the learned operator, not by the reference it is scored against [5, 6, 19]. Figure 3 shows the co-simulation energy defect vanishes linearly with the macro-step, the quantitative statement that the conservation gate (8) is always reachable — the property that distinguishes a disciplined co-simulation [14, 15] from an ad-hoc domain hand-off. Figure 4 shows the ensemble is calibrated at usable sizes and quantifies the $O(M^{-1/2})$ penalty inflation must offset, so the credible intervals the safety layer reads are earned rather than assumed.

8.2 Quench detection against the literature

The fused detector should be read against the published quench-detection literature, and specifically against the slow-NZPV problem that motivates it. Marchevsky’s review makes the core point quantitatively: in HTS conductors the large temperature margin and slow propagation mean a hot spot can reach damaging temperature before a measurable resistive voltage appears, so voltage detection alone is inadequate [26]. The community response has been non-voltage observables: Salazar *et al.* demonstrated fibre-Bragg thermometry on VIPER cable at SULTAN, reporting signal-to-noise ratios of 4–32 for heater-induced quench-like events with response times from effectively instantaneous to a few seconds [27]; Teyber *et al.* showed that current-distribution monitoring can detect quench and damage in fusion cables [28]. Our result is complementary and, by construction, evaluated in a harder regime: the ROC of table 2 is computed with per-sample SNRs set 10–30 $\times$ below the design-basis fault, precisely to avoid the saturated regime where every detector scores $\text{AUC} = 1$. In that weak-fault regime the fusion of an uncorrelated strain-rate and magnetization pair lifts the true-positive rate at 1% false alarm from 0.372/0.671 (single channels) to 0.857 — a large gain exactly where a protection trip is decided.

8.3 NV sensitivity against the quantum-sensing literature

The in-winding sensitivity of 0.189 pT Hz $^{-1/2}$ sits comfortably inside the envelope established for ensemble NV magnetometry. Degen, Reinhard and Cappellaro’s review frames the shot-noise scaling that equation (25) instantiates [24], and Barry *et al.* analyse the concrete route to broadband ensemble sensitivities in the picotesla-and-below range through improvements to T_2^* , readout fidelity and diamond material [25]. Our conservative parameter set ($N = 10^{12}$, $T_2^* = 1\text{ }\mu\text{s}$, $C = 0.03$) is well within what that literature reports, so the sensitivity claim does not rest on an aggressive device; the physically important observation is one of margin: a millitesla quench step is $\sim 10^9$ above the per-shot floor, so the precursor is single-shot detectable and the picotesla headroom is available for multiplexing many in-winding channels and for tolerating in-service degradation of T_2^* and C .

8.4 Relation to learned control and the Kronos series

The architecture is complementary to learned control. Degraeve *et al.* demonstrated a reinforcement-learning *policy* that commands coil voltages on TCV directly from magnetics, including negative-triangularity shapes [7]; the four-twin shadow is a *model* that runs ahead of the plant with explicit uncertainty, conservation and acceptance gates. The two compose: a learned policy, or the reference-governor model-predictive controller of the companion burn-control study [34], can act on the shadow’s predicted, uncertainty-annotated state rather than the raw measurement, inheriting the $\geq 50\text{ ms}$ lead and the calibrated coverage of equation (18), with the certified control-barrier safety filter [35] and the maturity-gated actuation-authority policy [36] beneath it as the last line of defence. The learned operators of §4.1 are developed in their own right — the equilibrium reconstruction robust to sensor dropout [32] and the differentiable flux surrogate with abstention monitoring [33] — the calibrated-uncertainty and conformal-abstention machinery in the companion on out-of-distribution authority hand-back [37], the state-estimation input to equations (15)–(17) in the graph-neural diagnostic fusion of the plasma diagnostic suite [38], and the magnet twin’s centrepost channel in the remaining-useful-life prognostics of the consumable centrepost. Relative to a purely learned controller, the contribution here is the conservation-respecting multi-domain coupling (§2.4, §3.2), the differentiable linearisation (equation 21), the real-time self-consistency monitor (equation 23), and the maturity-gated authority that keeps an immature twin advisory rather than commanding.

9 Limitations

Two honest caveats bound the claims. First, the magnet-twin quench results are twin-synthetic: every signal in the ROC study is numerically generated, and the confirming measurement is a coil-prototype hardware-in-the-loop bench, which is the gate to full-plant authority in the maturity ladder (§7.3); the numerical-formulation results of §3 are controlled convergence experiments on canonical test problems, chosen so that the order of accuracy can be measured exactly, and are verification of the scheme rather than validation against a device. Second, the in-winding NV sensor’s neutron tolerance is the one open engineering item: the analytical sensitivity of equation (25) assumes ensemble T_2^* and contrast that will degrade under the high background field, cryogenic operation, and neutron irradiation of the diamond (NV charge-state stability). The mitigation is threefold and is itself a registered work item (KFE-CF-SH-037): shielded siting, a replaceable sensor cartridge, and a model-fusion backup that retains precursor $\text{AUC} \geq 0.95$ with

the NV channel degraded by up to $39\times$ — so the protective function survives even a heavily degraded sensor. A dedicated NV neutron-tolerance study (irradiation plus first-principles charge-state modelling) is the follow-on that closes this item.

10 Conclusion

The Kronos control architecture is a four-twin shadow that runs ahead of the plant across every coupled domain: MHD-stability, thermomechanics, neutronics and power-systems twins, advanced by a conserving multi-rate co-simulation whose interface is a waveform-relaxation fixed point (equations 6–7) and whose energy and flux gates (equations 8–9) we show close with the macro-step; made predictive by learned flux and equilibrium neural operators that the controller linearises on the fly (equation 21) and whose equilibrium reference we discretise to demonstrated second order (figure 2); carrying calibrated, coverage-tested uncertainty through an ensemble-Kalman assimilation (equations 15–17, figure 4); audited each tick by a real-time self-consistency monitor (equation 23); and admitted to control authority only through explicit, pre-registered acceptance inequalities (§4.3). Its magnet twin is demonstrated in full at twin-synthetic fidelity: a fused strain-rate-plus-magnetization quench-precursor detector at AUC 0.992 and true-positive rate 0.857 at 1% false alarm (table 2), and an in-winding NV-diamond magnetometer at $0.189 \text{ pTHz}^{-1/2}$ with per-shot SNR of order 10^9 (table 3), with the breeder centrepost (16.84 T) and burner plug (26.49 T) modelled as the distinct magnets they are. The demonstrated subsystem and the demonstrated numerics provide the proven pattern the remaining twins follow, and the maturity-gated authority argument makes the route to a fully autonomous compact ST breeder a defined and testable sequence.

Acknowledgments

Part of the Kronos Fusion Energy 2026 design series. The author thanks the Kronos physics de-risking programme for the frozen result set underlying every quantitative claim, and acknowledges the NVIDIA GPU HPC campaign on which the learned-operator training and ensemble inference were run.

Reproducibility. This paper is archived at doi:10.5281/zenodo.22645805 and at its official page, <https://www.kronosfusionenergy.com/publications/four-coupled-digital-twins-for-a-compact-spherical-toka.html>. The magnet-twin detection metrics are frozen anchors in the Kronos physics de-risking register [30] (doi:10.5281/zenodo.22133057): the fused-detector ROC (KX-L1-A20), the NV-diamond sensor model (KX-L1-A22), and the magnet-twin margin map (KX-L1-A15). The numerical-formulation experiments (Grad-Shafranov convergence, figure 2; multi-rate energy defect, figure 3; ensemble coverage, figure 4) are regenerable from the single-file figure-generation script deposited with this paper, with fixed random seeds, on commodity hardware.

Data availability

The ROC curves, the NV sensor parameters, the magnet-twin load-line map, and the convergence and coverage data used for the figures are archived with the deposit at doi:10.5281/zenodo.22645805. All quantitative values trace to the register [30] by the serial identifiers cited inline. Any economic analysis is reported separately and is not part of the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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